1. Which of the following orbits around the Sun?

A. Planets
B. Comets
C. Asteroids
D. All of the above

---

2. As a nebula shrinks under the influence of gravity, it

A. spins slower
B. spins faster
C. loses its spin
D. spins into a protosun

---

3. Strictly speaking, in every second that passes, the mass of the Sun

A. decreases
B. remains constant
C. increases
D. reinvents itself

---

4. Which planet is more dense than water?

A. Mars
B. Venus
C. Neptune
D. All of the above

---

5. During the time of a new Moon, the

A. Sun is between Earth and the Moon
B. Moon is between the Sun and Earth
C. Earth is between the Sun and the Moon
D. None of the above

---

6. During the time of a full Moon, the

A. Sun is between Earth and the Moon
B. Moon is between the Sun and Earth
C. Earth is between the Sun and the Moon
D. None of the above

---

7. Asteroids are small rocky bodies that

A. orbit the Sun
B. mainly reside between Mars and Jupiter
C. are smaller than Earth's Moon
D. All of the above

---

8. The tails of comets point in a direction

A. toward the Sun
B. away from the Sun
C. at nearly right angles to the Sun
D. None of the above

---

9. Which of these makes contact with Earth's surface?

A. Meteor
B. Meteorite
C. Meteoroid
D. None of the above

---

10. Knowing the names of the constellations tells us much about the

A. stars that comprise them
B. people in the cultures that named them
C. difference between stars and planets
D. All of the above

---

11. Which of these stars radiates light of the longest wavelength?

A. Red star
B. Yellow star
C. Blue star
D. Violet star

---

12. We measure the Sun's luminosity as (L_{\text{Sun}}).

If we were on a spaceship twice as far away from the Sun, its apparent brightness would appear
A. the same
B. half as much
C. one quarter as much
D. four times as much

---

13. On the H–R diagram, the Sun is

A. an average star
B. seen to be special
C. a low-temperature star
D. especially bright

---

14. A dying star that has collapsed to a small size and is cooling off would appear in which part of the H–R diagram?

A. Lower left
B. Upper left
C. Lower right
D. Upper right

---

15. The source of energy in the Sun and stars is

A. chemical reactions
B. thermonuclear reactions
C. Both of the above
D. None of the above

---

16. When a star collapses to one-tenth size, gravitation at its surface becomes

A. one tenth as much
B. the same
C. 10 times as much
D. 100 times as much

---

17. When a giant star collapses to become a black hole, gravity is greatly increased

A. at its surface
B. at its center
C. in all surrounding space
D. All of the above

---

18. If the Sun collapsed to become a black hole, the orbit of Earth would

A. remain unchanged
B. be pulled inward toward the black hole
C. spiral outward away from the black hole
D. be a straight-line path

---

19. The motion of individual stars in a galaxy normally follow

A. elliptical orbits around the center of the galaxy
B. completely random paths
C. straight lines for the most part
D. circular orbits around the center of the galaxy

---

20. If the distance between your eye and a lamp is halved,

A. the light will appear to be two times brighter
B. the light will appear to be two times dimmer
C. the light will appear to be four times brighter
D. the light will appear to be eight times dimmer

---

21. How will the spectrum of a star moving toward us change?

A. The wavelength will be stretched (redshift)
B. The wavelength will be compressed (blueshift)

---

22. Hubble's law has to do with the

A. expansion of the universe
B. steady state of the universe
C. contraction of the universe
D. cosmic background radiation

---

23. Scientific evidence supports the date of the Big Bang event at

A. 7,000 years ago
B. 100,000 years ago
C. 5 billion years ago
D. 13.7 billion years ago

---

24. Which scenario do you think is more likely?

A. Heat Death
B. The Big Rip
C. Other
D. We really don't know

Answer Key

1. D
2. B
3. A
4. D
5. B
6. C
7. D
8. B
9. B
10. B
11. A
12. C
13. A
14. A
15. B
16. D
17. A
18. A
19. A
20. C
21. B
22. A
23. D
24. D